R&D | RetailPulse

Document ID: RP-EMF-004 | Version: 2.0 | Date: 2026-10-21 | Lane: Insights Analyst

Forecast Error Metric Framework

> **In simple words**
>
> This document defines the math rules for error: what the baseline is, how WAPE, Bias and units are calculated, and when a family counts as high error.

1. Baseline forecasts (simple on purpose)

Baseline	Rule	Role
A: Same weekday last week	Forecast for a day = actual sales 7 days before (same store, same family)	Main baseline
B: 4-week same-weekday average	Average of actual sales 7, 14, 21 and 28 days before	Comparison baseline

Both use only past data. The first 28 days have no baseline and are excluded. Always check that the earlier date is exactly 7 (14, 21, 28) days before, because some dates are missing.

2. Metrics

Metric	Formula	Plain meaning
Absolute error	ABS(actual - forecast)	Size of the miss on one row
WAPE (main metric)	SUM(abs error) / SUM(actual)	Error as a share of sales. Works with zero-sales rows.
Accuracy %	1 - WAPE	How close we are
Bias %	(SUM forecast - SUM actual) / SUM actual	Plus = guessing too high. Minus = too low.
Excess units	SUM of MAX(forecast - actual, 0)	Units we forecast but did not sell
Shortfall units	SUM of MAX(actual - forecast, 0)	Units sold above forecast (risk of empty shelf)
Error share %	Group abs error / total abs error	How much of the total problem this group causes
MAE (optional)	AVERAGE of abs error	Average miss in units
MAPE (optional)	Average of abs error / actual	Fails when actual = 0. Avoid as main metric.

> **Never average WAPE values**
> WAPE for any group = SUM of absolute error for that group / SUM of actual sales for that group. Averaging WAPE values from smaller groups gives a wrong answer.

3. Worked example (example numbers only)

Month	Abs error	Actual sales	WAPE
Month 1	100	500	20%
Month 2	200	1,000	20%
Month 3	300	500	60%
Total	**600**	**2,000**	**30%**

The correct total WAPE is 600 / 2,000 = 30%. The average of 20%, 20% and 60% is 33.3%, which is wrong.

4. Why we show both error and bias

A family can have a small Bias % but a big WAPE. This happens when over-forecasts and under-forecasts cancel each other. Bias hides the problem. WAPE shows the size of the miss. Bias shows the direction. Always report both.

5. High-error rule

> **Starting rule**
>
> A family is **High Error** if its WAPE is above **30%** and its total sales are above a minimum volume. Rank families by **Error share %** to find the biggest problems.

- 30% is a starting point. After the first run, look at the WAPE spread and adjust.
- Ignore tiny-volume families when flagging, because WAPE is unstable for them. Report them separately.
- Priority = high WAPE **and** high Error share. A high-WAPE family with tiny sales matters less.

6. Reporting levels

Level	Used for
Overall	Executive KPIs
Family	Category ranking, Pareto
State, city, store type, cluster	Region analysis
Month, quarter, named period, weekday	Season analysis
Holiday, payday, promotion flag	Event analysis

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